

ECM Coolant Temperature Gauge Output Signal Malfunction

Symptoms

- ECM coolant temperature gauge output signal is **not** available.

How To Use This Tree

This symptom tree can be used to troubleshoot an ECM coolant temperature gauge output signal malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check Customer Interface Box Wiring	
	STEP 1A. Check Coolant Temperature Meter Signal Wire for Open	Less than 10 ohms?
	STEP 1B. Check Coolant Temperature Meter Signal Wire for Wire to Wire Short	Less than 10 ohms?
	STEP 1C. Check Coolant Temperature Meter Signal Wire for Short to Ground	Less than 10 ohms?
STEP 2.	Check Engine Harness to Customer Interface Box Cable	
	STEP 2A. Check Coolant Temperature Meter Signal Wire for Open	Less than 10 ohms?
	STEP 2B. Check Coolant Temperature Meter Signal Wire for Wire to Wire Short	Less than 10 ohms?

STEP 1. Check Customer Interface Box Wiring

STEP 1A. Check Coolant Temperature Meter Signal Wire for Open

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the coolant temperature meter signal wire for an open. • Place one test lead on the coolant temperature meter signal pin in connector C2. • Place the other test lead on coolant temperature meter signal terminal on the X4 connector.	Less than 10 ohms? YES	1B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete

STEP 1B. Check Coolant Temperature Meter Signal Wire for Wire to Wire Short

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
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Check the coolant temperature meter signal wire for wire to wire short. - Place one test lead on the coolant temperature meter signal pin in connector C2. - Place the other test lead on each of the remaining terminals in X4 connector.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete
	Less than 10 ohms? NO	1C

STEP 1C. Check Coolant Temperature Meter Signal Wire for Short to Ground

Conditions :

- Open the customer interface box
- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box.

Action	Specification/Repair	Next Step
Check the coolant temperature meter signal wire for short to ground. - Place one test lead on the coolant temperature meter signal pin in connector C2. - Place the other test lead on panel ground.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html).	Repair complete
	Less than 10 ohms? NO	2A

STEP 2. Check Engine Harness to Customer Interface Box Cable

STEP 2A. Check Coolant Temperature Meter Signal Wire for Open

Conditions :

- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step
Check coolant temperature meter signal wire for an open. • Place a jumper between the coolant temperature meter signal pin and the droop adjust potentiometer signal pin in the C9 connector. • Place one test lead in the coolant temperature meter signal pin of the C2 connector. • Place the other test lead in the droop adjust potentiometer signal pin of the C2 connector.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the cable.	Repair complete

STEP 2B. Check Coolant Temperature Meter Signal Wire for Wire to Wire Short**Conditions :**

- Disconnect customer interface box to engine harness cable connector C2 from the customer interface box
- Disconnect customer interface box to engine harness cable connector C9 from the engine harness.

Action	Specification/Repair	Next Step
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	Less than 10 ohms? YES	Repair complete
	Less than 10 ohms? NO Repair : Refer to Section TF in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail System Series Engines, Bulletin 4021493 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021493.html), or the Troubleshooting and Repair Manual, Electronic Control System, QSK38, QSK50, and QSK60, CM850 Modular Common Rail System Series Engines, Bulletin 4021533 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021533.html), or refer to the OEM service manual.	Repair complete
Check coolant temperature meter signal wire for a wire to wire short. - Place one test lead in the coolant temperature gauge output signal pin in connector C2. - Place the other test lead on each of the remaining pins in the C2 connector.		